

HR ANALYTICS USING SQL

PROJECT OBJECTIVES:

- Analyze employee attrition patterns
- Identify high-risk employee segments
- Investigate retention drivers
- Generate actionable workforce insights

DATASET:

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

TOOLS USED:

- MYSQL
- EXCEL

GITHUB:

<https://github.com/ramitsakhuja/HR-Analytics-SQL-PowerBI.git>

CREATED BY:

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-- 1. Total Employees

```
SELECT
    COUNT(*) AS Total_Employees
FROM
    employee_attrition;
```

-- 2. Attrition Count

```
SELECT
    COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes';
```

-- 3. Attrition Rate

```
SELECT
    ROUND(SUM(CASE
        WHEN Attrition = 'Yes' THEN 1
        ELSE 0
    END) * 100.0 / COUNT(*),
    2) AS Attrition_Rate
FROM
    employee_attrition;
```

-- 4. Average Employee Age

```
SELECT
    ROUND(AVG(Age), 2) AS Average_Age
FROM
    employee_attrition;
```

-- 5. Average Monthly Income

```
SELECT
    ROUND(AVG(MonthlyIncome), 2) AS Average_Monthly_Income
FROM
    employee_attrition;
```

-- 6. Employee Distribution by Department

```
SELECT
    Department, COUNT(*) AS Employee_Count
FROM
    employee_attrition
GROUP BY Department
ORDER BY Employee_Count DESC;
```

-- 7. Attrition by Department

```
SELECT
    Department, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY Department
ORDER BY Attrition_Count DESC;
```

-- 8. Average Salary by Department

```
SELECT
    Department, ROUND(AVG(MonthlyIncome), 2) AS Avg_Salary
FROM
    employee_attrition
GROUP BY Department
ORDER BY Avg_Salary DESC;
```

-- 9. Employee Distribution by Gender

```
SELECT
    Gender, COUNT(*) AS Employee_Count
FROM
    employee_attrition
GROUP BY Gender;
```

-- 10. Attrition by Gender

```
SELECT
    Gender, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY Gender
ORDER BY Attrition_Count DESC;
```

-- 11. Attrition by Marital Status

```
SELECT
    MaritalStatus, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY MaritalStatus
ORDER BY Attrition_Count DESC;
```

-- 12. Attrition by Education Field

```
SELECT
    EducationField, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY EducationField
ORDER BY Attrition_Count DESC;
```

-- 13. Attrition by Job Role

```
SELECT
    JobRole, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY JobRole
ORDER BY Attrition_Count DESC;
```

-- 14. Attrition by Overtime

```
SELECT
    OverTime, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY OverTime;
```

-- 15. Attrition by Job Satisfaction

```
SELECT
    JobSatisfaction, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```

-- 16. Attrition by Environment Satisfaction

```
SELECT
    EnvironmentSatisfaction, COUNT(*) AS Attrition_Count
FROM
```

```
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY EnvironmentSatisfaction
ORDER BY EnvironmentSatisfaction;
```

-- 17. Attrition by Work Life Balance

```
SELECT
    WorkLifeBalance, COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY WorkLifeBalance
ORDER BY WorkLifeBalance;
```

-- 18. Tenure Band Analysis

```
SELECT
    CASE
        WHEN YearsAtCompany <= 2 THEN '0-2 Years'
        WHEN YearsAtCompany <= 5 THEN '3-5 Years'
        WHEN YearsAtCompany <= 10 THEN '6-10 Years'
        ELSE '11+ Years'
    END AS Tenure_Band,
    COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
GROUP BY Tenure_Band
ORDER BY Attrition_Count DESC;
```

-- 19. Salary Band Analysis

```
SELECT
    CASE
        WHEN MonthlyIncome < 5000 THEN 'Low Income'
        WHEN MonthlyIncome < 10000 THEN 'Medium Income'
        ELSE 'High Income'
    END AS Salary_Band,
    COUNT(*) AS Attrition_Count
FROM
    employee_attrition
WHERE
    Attrition = 'Yes'
```

```
GROUP BY Salary_Band
ORDER BY Attrition_Count DESC;
```

-- 20. Top 5 Highest Paying Job Roles

```
SELECT
    JobRole, ROUND(AVG(MonthlyIncome), 2) AS Avg_Salary
FROM
    employee_attrition
GROUP BY JobRole
ORDER BY Avg_Salary DESC
LIMIT 5;
```

-- 21. Department-wise Attrition Rate

```
SELECT
    Department,
    COUNT(CASE
        WHEN Attrition = 'Yes' THEN 1
    END) AS Attrition_Count,
    COUNT(*) AS Total_Employees,
    ROUND(COUNT(CASE
        WHEN Attrition = 'Yes' THEN 1
    END) * 100.0 / COUNT(*),
        2) AS Attrition_Rate
FROM
    employee_attrition
GROUP BY Department
ORDER BY Attrition_Rate DESC;
```

-- 22. Rank Job Roles by Attrition

```
SELECT
    JobRole,
    COUNT(*) AS Attrition_Count,
    RANK() OVER(
        ORDER BY COUNT(*) DESC
    ) AS Attrition_Rank
FROM
    employee_attrition
WHERE
    Attrition='Yes'
GROUP BY JobRole;
```

KEY BUSINESS INSIGHTS:

- Attrition Rate: 16.12%
- Research & Development records the highest employee attrition.
- Laboratory Technicians experience the highest attrition among job roles.
- Employees with 0–2 years tenure are most likely to leave.
- Overtime employees show significantly higher attrition.
- Low-income employees account for the highest employee exits.
- Environment satisfaction is a major retention driver.

SKILLS DEMONSTRATED:

- SQL Aggregations
- CASE Statements
- GROUP BY & ORDER BY
- KPI Analysis
- Window Functions
- Business Intelligence
- Data Storytelling

THANK YOU